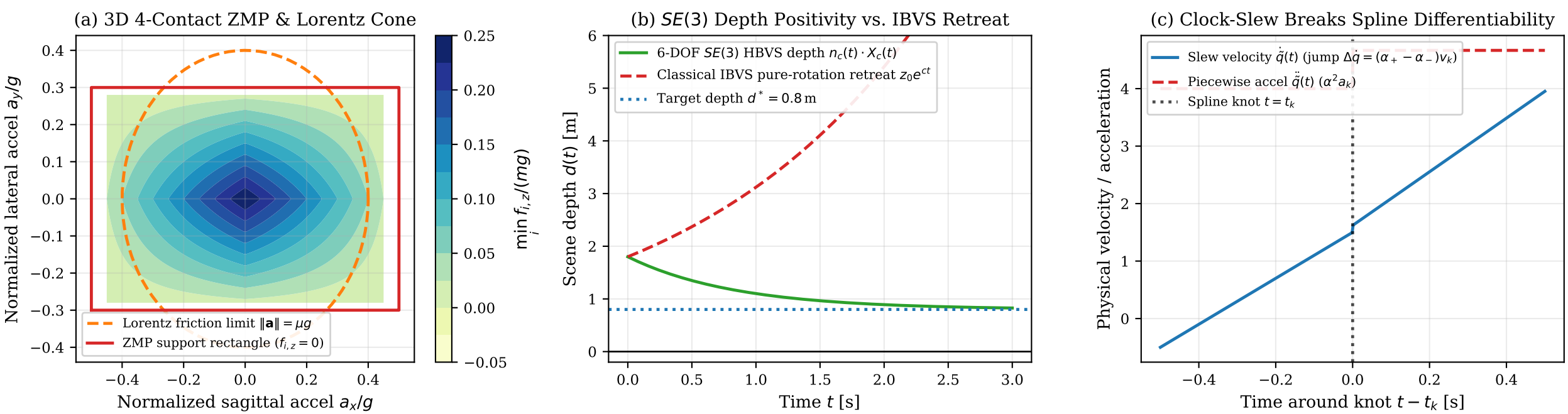

(a) 3D 4-Contact ZMP & Lorentz Cone
Normalized lateral accel a_y/g
Normalized sagittal accel a_x/g
Lorentz friction limit $\|\mathbf{a}\| = \mu g$
ZMP support rectangle ($f_{i,z} = 0$)
$\min_i f_{i,z}/(mg)$
(b) $SE(3)$ Depth Positivity vs. IBVS Retreat
Scene depth $d(t)$ [m]
Time t [s]
6-DOF $SE(3)$ HBVS depth $n_c(t) \cdot X_c(t)$
Classical IBVS pure-rotation retreat $z_0 e^{ct}$
Target depth $d^* = 0.8$ m
(c) Clock-Slew Breaks Spline Differentiability
Physical velocity / acceleration
Time around knot $t - t_k$ [s]
Slew velocity $\dot{\tilde{q}}(t)$ (jump $\Delta\dot{q} = (\alpha_+ - \alpha_-)v_k$)
Piecewise accel $\ddot{\tilde{q}}(t)$ ($\alpha^2 a_k$)
Spline knot $t = t_k$